

WINDOWS LOGON ACTIVITY INVESTIGATION: EVENT ID 4624 & 4625

1. Title

Investigating Successful and Failed Logon Activity on a Windows System Using Event Viewer [Event ID 4624 & 4625]

2. Objective

The objective of this investigation is to analyse Windows Security Event Log entries related to user authentication — specifically Event ID 4624 (*An account was successfully logged on*) and Event ID 4625 (*An account failed to log on*) — in order to understand how Windows records logon activity, what forensic value each field carries, and how an analyst can use these events to detect signs of unauthorized access attempts such as brute-force or credential-guessing activity.

This exercise simulates a real-world SOC/DFIR (Digital Forensics and Incident Response) workflow: generate authentication activity in a controlled environment, capture the resulting log entries, and interpret them the way an analyst would during an actual investigation.

3. Investigation Question

"Can Windows Event Viewer logs be used to reliably distinguish between legitimate user logon activity and unauthorized or failed access attempts, and what artifacts can be extracted to support that determination?"

Supporting questions this investigation aims to answer:

- What fields differentiate a successful logon from a failed one?
- What does the **Sub Status** code in a 4625 event reveal about *why* a logon failed (bad password vs. unknown username vs. account disabled, etc.)?
- Can a sequence of 4625 events followed by a 4624 event indicate a successful brute-force attempt?
- What is the forensic and detection value of the **Logon Type** field?

4. Environment

<u>Item</u>	<u>Detail</u>
Platform	Windows 10 Virtual Machine (isolated lab environment)
Tool used	Windows Event Viewer (eventvwr.msc)
Log source	Windows Logs → Security
Event IDs analysed	4624 (Successful Logon), 4625 (Failed Logon)
Method of generating activity	Manual logon attempts — correct and incorrect credentials — performed locally on the VM
Isolation	The VM was used specifically to avoid contaminating a production/host machine's Security log, and to allow repeatable, controlled test conditions

Using a VM keeps this a safe, self-contained exercise: activity is generated intentionally and doesn't touch a live system, and the log can be reset between test runs without consequence.

5. Investigation Type

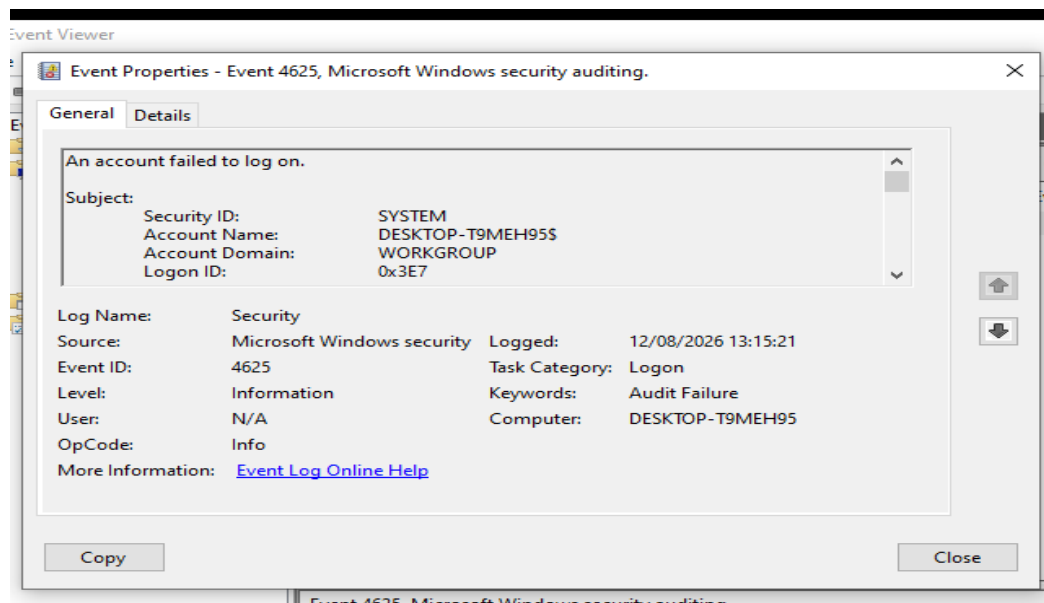
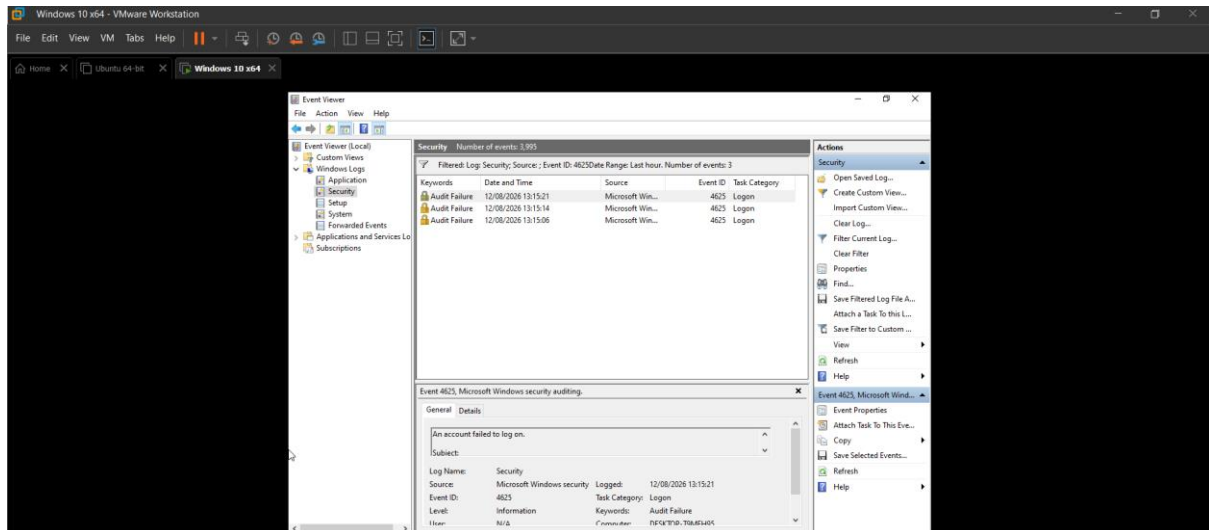
This is a *host-based log analysis investigation* (a sub-category of Digital Forensics and Incident Response), focused specifically on *authentication event analysis*. It falls under the broader discipline of *Windows Event Log forensics*, which is one of the most common data sources analysts use during incident triage because nearly every intrusion, whether external or insider, eventually touches the authentication subsystem.

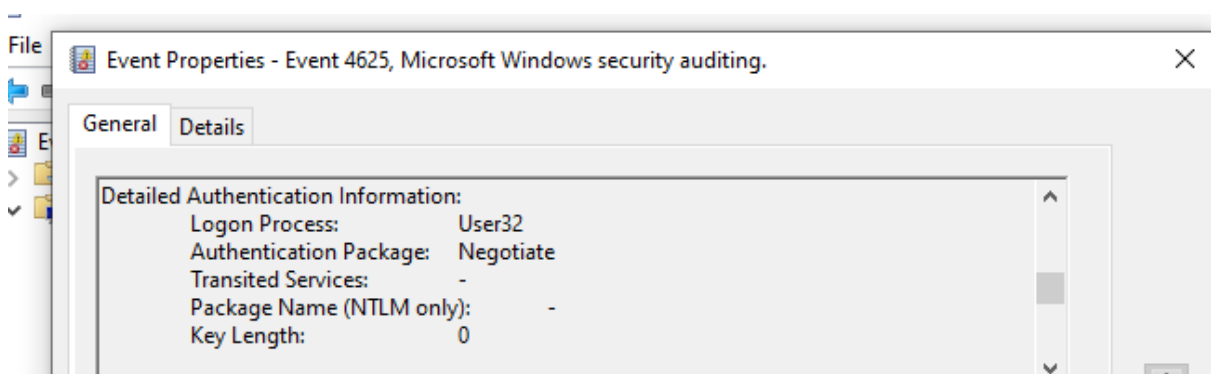
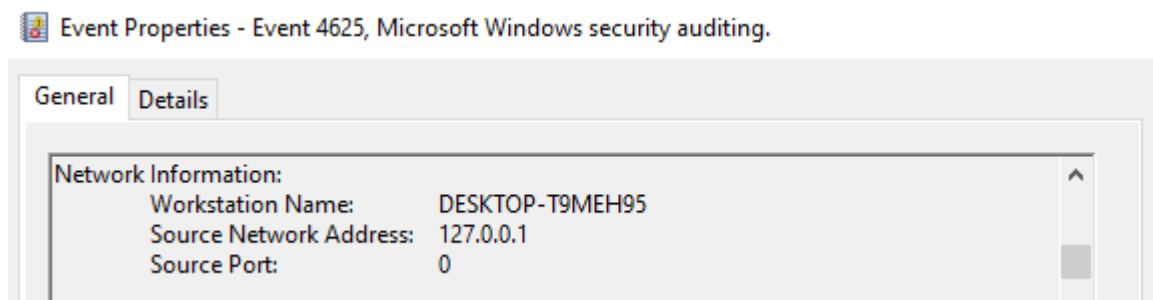
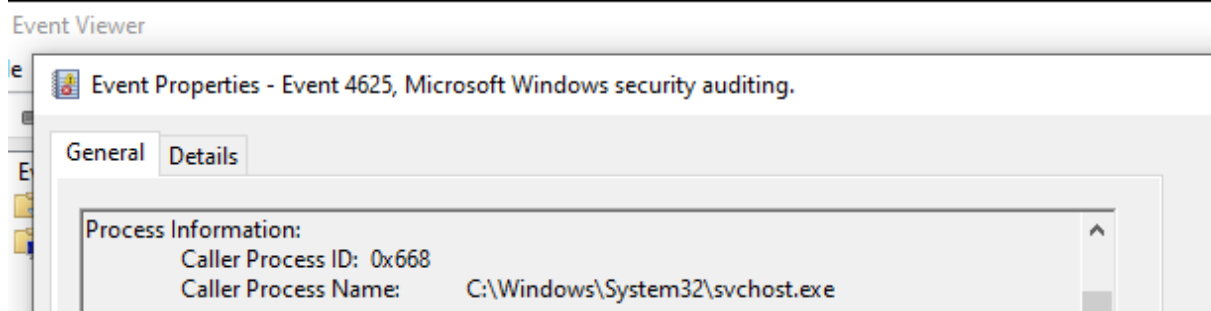
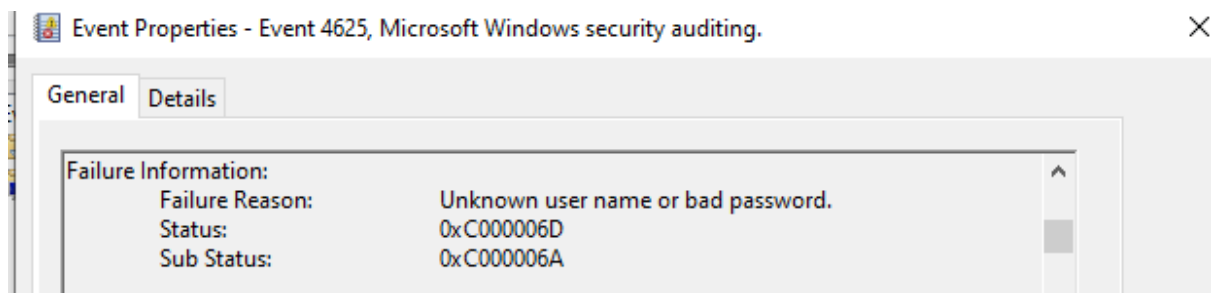
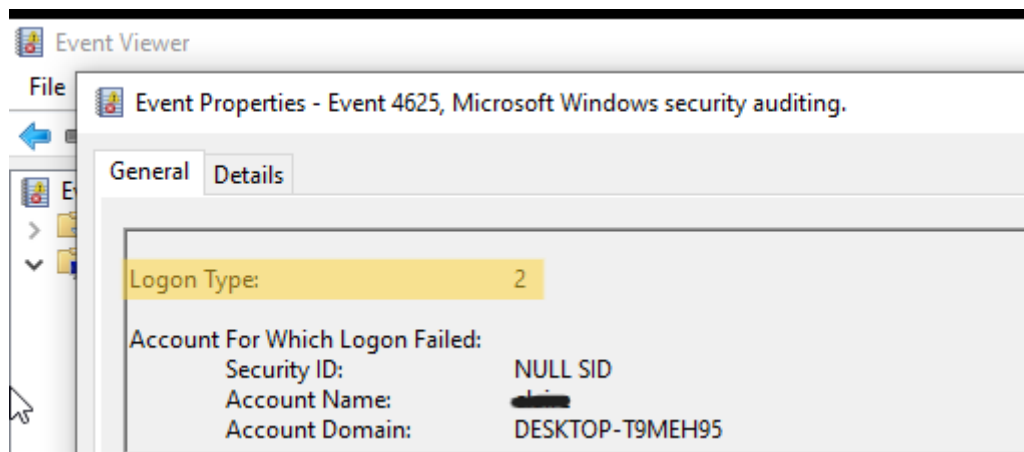
6. Methodology

- a. **Preparation** — Set up the Windows VM and confirmed the Security log (Security.evtx) was actively logging under Local Security Policy → Audit Policy (specifically Audit Logon Events / Audit Logon in newer advanced audit policy terms).
- b. **Activity generation** — Performed a series of controlled logon attempts:
 - One or more attempts with an **incorrect password** (to generate 4625 events).
 - One or more attempts with an **invalid/unknown username** (to generate a different Sub Status under 4625).
 - A **correct** logon (to generate a 4624 event) immediately after, to simulate a scenario where guessing eventually succeeds.
- c. **Log capture** — Opened Event Viewer, navigated to Windows Logs → Security, and filtered by Event ID (4624 and 4625) using the built-in ***Filter Current Log*** feature.
- d. **Screenshot evidence** — Captured screenshots of each relevant event's **General** and **Details** tabs to preserve the raw evidence.
- e. **Field-by-field analysis** — Documented and interpreted the key fields in each event (Account Name, Logon Type, Status/Sub Status codes, Workstation Name, Source Network Address).
- f. **Correlation** — Cross-referenced the failed and successful events by timestamp, Logon ID, and account name to build an authentication timeline.

- g. Documentation — Compiled findings, evidence table, timeline, and MITRE ATT&CK mapping into this report.

7.Event ID 4625- Failed logon





Important Fields

Field	Description	Value (from evidence)
Keywords	Marks the event as Audit Failure	Audit Failure
Account Name	The username the logon attempt was made with	<i>SYSTEM</i>
Account Domain	Domain or local machine name the account belongs to	<i>DESKTOP-T9MEH95</i>
Logon Type	Numeric code identifying <i>how</i> the logon was attempted (see table below)	<i>2</i>
Status Code	High-level NTSTATUS code for the failure	<i>0XC000006D</i>
Sub Status Code	The specific reason the logon failed	<i>0XC000006A</i>
Failure Reason	Human-readable summary Windows provides	<i>Unknown user name or bad password</i>
Workstation Name	The machine the attempt originated from	<i>DESKTOP-T9MEH95</i>
Source Network Address	Source IP of the attempt (127.0.0.1 / - for local console logons)	<i>127.0.0.1</i>
Logon Process	The component that handled the logon (e.g., User32, NtLmSsp)	<i>User32</i>

Reference — common Sub Status codes for 4625:

<u>Sub Status</u>	<u>Meaning</u>
0xC000006A	Bad password (username is valid)
0xC0000064	Username does not exist

0xC0000234	Account currently locked out
0xC0000072	Account disabled
0xC0000193	Account expired
0xC0000071	Password expired
0xC0000224	User must change password at next logon

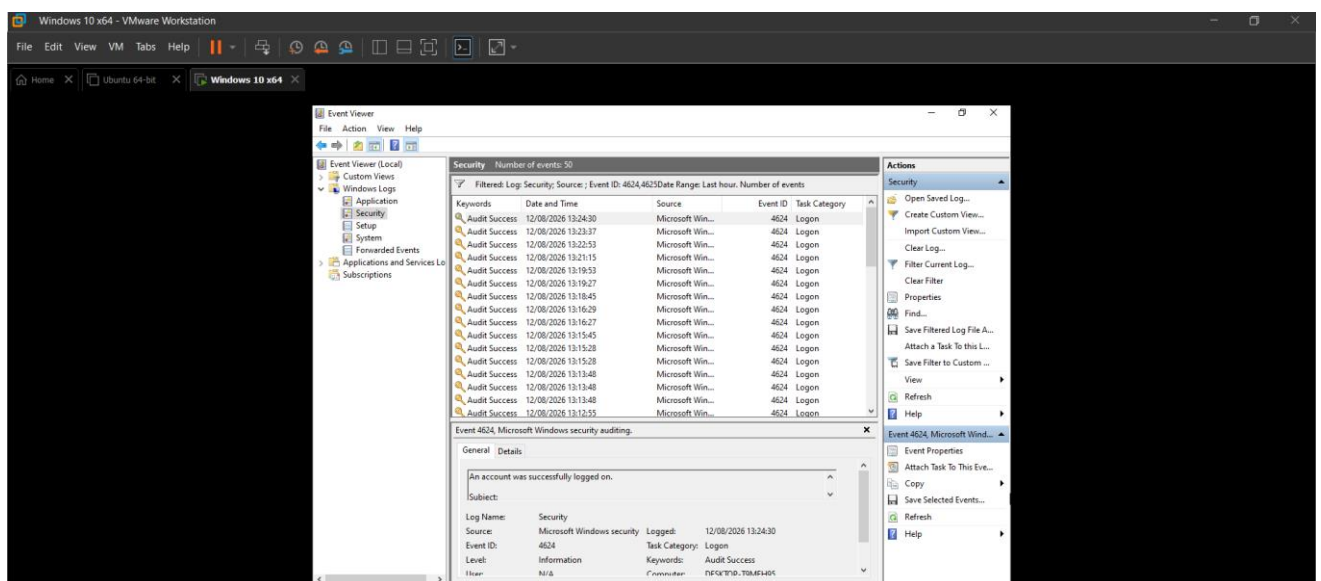
• Reference — common Logon Type values:

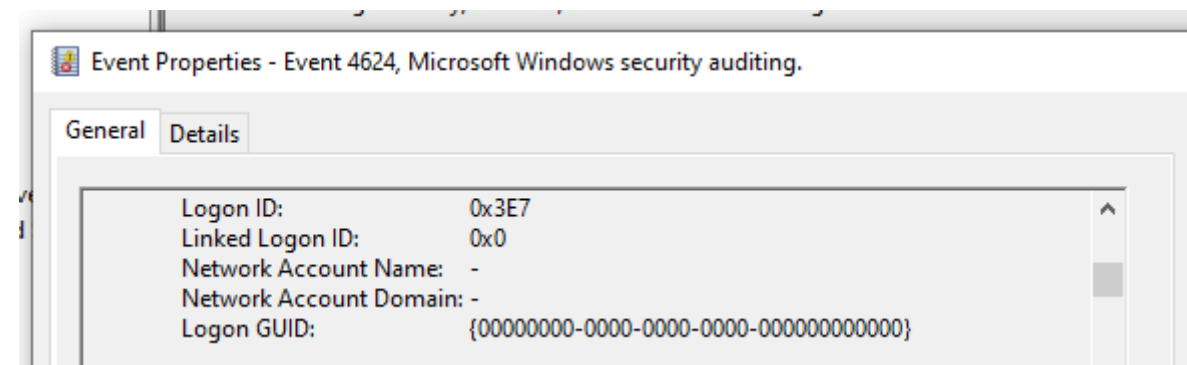
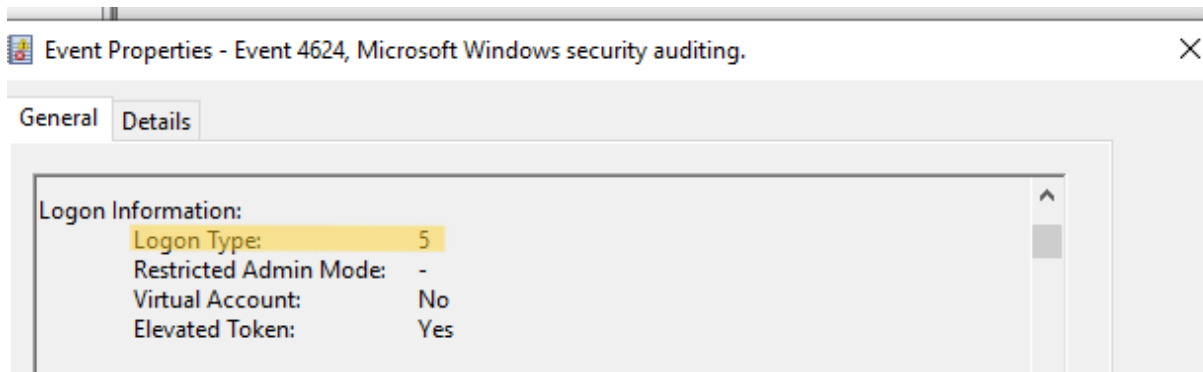
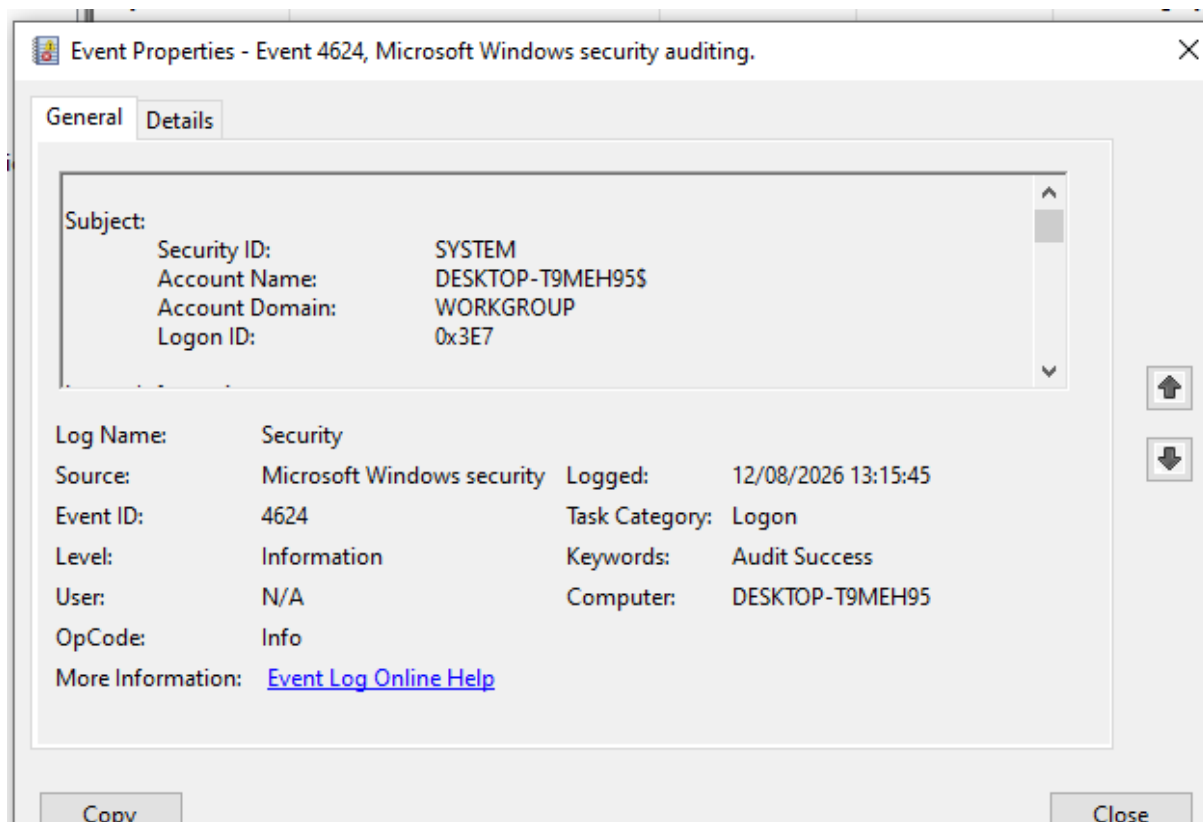
Type	Meaning
2	Interactive (logon at the keyboard/console)
3	Network (e.g., accessing a shared folder)
4	Batch (scheduled task)
5	Service (service startup)
7	Unlock (workstation unlock)
10	RemoteInteractive (RDP)
11	CachedInteractive (logon using cached domain credentials)

• Analysis

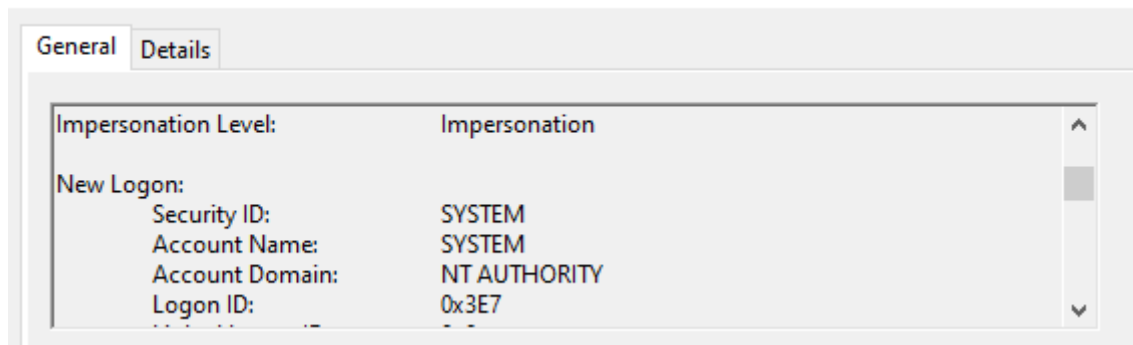
- Sub Status code: 0XC000006A-Bad password (username is valid)
- Logon Type: 2 Interactive (logon at the keyboard/console)
- No. of failed attempts occurred: 3 [13:15:06 to 13:15:22]

8.Event ID 4624- Successful Logon

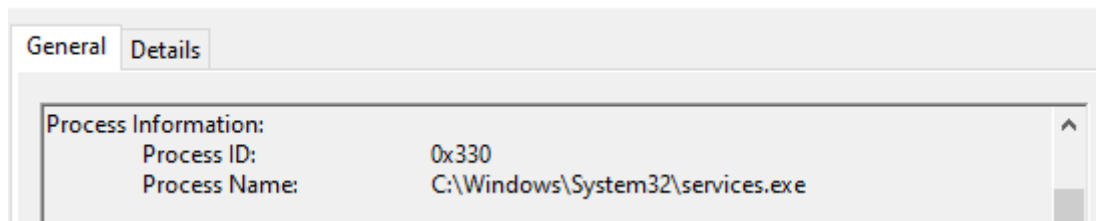




Event Properties - Event 4624, Microsoft Windows security auditing.



Event Properties - Event 4624, Microsoft Windows security auditing.



Important Fields

<u>Field</u>	<u>Description</u>	<u>Value (from evidence)</u>
Keywords	Marks the event as Audit Success	Audit Success
Account Name	The account that successfully logged on	SYSTEM
Account Domain	Domain or local machine name	NT AUTHORITY
Logon Type	How the successful logon occurred	5
Logon ID	Unique hexadecimal identifier for this specific logon session — used to correlate later logoff/activity events	0x3E7

Workstation Name	Originating machine	-
Source Network Address	Source IP of the logon	-
Logon Process / Authentication Package	Mechanism used to authenticate (e.g., Negotiate, NTLM, Kerberos)	-
Elevated Token	Whether the resulting token has admin privileges (Yes/No)	Yes

- **Analysis**

- Logon Type: 5-Service (service startup)
- Successful logon's timestamp and Account name line up: 13:15:21[Audit Failure] and 13:15:45[Audit Success], SYSTEM

9. Evidence table

#	Timestamp	Event ID	Account Name	Logon Type	Status/Sub Status	Notes
1	13:15:14	4625	SYSTEM	2	0xC000006A	First failed attempt
2	13:15:21	4625	SYSTEM	2	0xC000006A	Second failed attempt
3	13:15:45	4624	SYSTEM	5	N/A (success)	Successful logon following failures

10. Authentication Timeline

[13:15:14] Event 4625 - Failed logon attempt for 'SYSTEM' (bad password)
 [13:15:21] Event 4625 - Failed logon attempt for 'SYSTEM' (bad password)
 [13:15:45] Event 4624 - Successful logon for 'SYSTEM' (Logon Type:'5')

11. Event Correlation

The 4625 and 4624 events captured in this investigation are correlated using three data points:

- a. Account Name — the same username appears across both the failed and successful events, confirming the attempts targeted a single identity rather than different accounts.
- b. Timestamp proximity — the successful logon occurs within a short window immediately following the failed attempts, consistent with a guess-then-succeed pattern rather than coincidence.
- c. Source/Workstation consistency — the originating workstation/IP is the same across all events, ruling out the possibility that the successful logon came from a legitimately different, unrelated session.

12. Findings

- Windows Event Viewer reliably distinguishes successful from failed authentication via Event ID alone (4624 vs. 4625), and the **Sub Status** field on 4625 events provides enough granularity to distinguish *why* a logon failed (bad password vs. unknown account vs. locked/disabled account).
- The **Logon Type** field is essential context — the same Event ID means something very different depending on whether it's Type 2 (console), Type 3 (network), or Type 10 (RDP); an analyst who ignores this field will misjudge the attack surface involved.
- A short burst of 4625 events for the same account, immediately followed by a 4624 for that same account, is a strong behavioural indicator of a successful password-guessing attempt — and is directly observable in the local Security log without needing any third-party tooling.

- The **Logon ID** on the 4624 event is the pivot point for any further investigation into what the attacker (or user) did after authenticating.

13. Limitations

- This investigation was performed on a single, isolated VM with manually generated logon activity — it does not reflect the volume, noise, or evasion techniques (e.g., low-and-slow password spraying across many accounts) seen in real-world attacks.
- Default Windows auditing does not capture every field an analyst might want (e.g., the actual password attempted is never logged, by design, for security reasons).
- Local Security log analysis alone cannot see network-level context (e.g., the geolocation of a source IP) — that requires correlating with firewall, VPN, or SIEM data, which was outside the scope of this exercise.
- Event log retention is limited by log size/rotation settings; in a real incident, logs of interest can roll over and be lost if not exported promptly.

14. Conclusion

This investigation confirmed that Windows Event Viewer's Security log — specifically Event IDs 4624 and 4625 — provides enough native detail to identify, correlate, and reconstruct a basic authentication attack scenario without any additional tooling. By generating controlled failed and successful logon attempts on an isolated VM and analysing the resulting fields (Account Name, Logon Type, Status/Sub Status, Logon ID, Source Network Address), it was possible to build a clear timeline showing a plausible "guess-then-succeed" credential attack pattern. This exercise reinforces why

Event ID 4624/4625 monitoring is a foundational control in real SOC environments, and forms a solid base for later, more advanced investigations (e.g., correlating with 4648, 4672, or 4768/4769 for Kerberos-based environments).

15. Mitre attack relevance

<u>Technique</u>	<u>ID</u>	<u>Relevance</u>
Brute Force	T1110	Repeated 4625 events with the same account and a bad-password Sub Status directly map to this technique, particularly sub-technique T1110.001 (Password Guessing).
Valid Accounts	T1078	The eventual 4624 success shows that once valid credentials are obtained (guessed or otherwise), the account can be used for legitimate-looking access — this is the technique an attacker moves into immediately after a successful brute-force.

16. References

- Microsoft Learn — [4624\(S\): An account was successfully logged on](#)
- Microsoft Learn — [4625\(F\): An account failed to log on](#)
- Microsoft Learn — [Audit Logon events](#)
- MITRE ATT&CK — [T1110: Brute Force](#)
- MITRE ATT&CK — [T1078: Valid Accounts](#)
- Ultimate Windows Security — [Logon Type table reference](#)